$T: \mathbb{R}^2 \to \mathbb{R}^2 \quad T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3x \\ -y \end{bmatrix}$

Pergunta: Verifique se essa transformação é linear.

Passo 1: Testar origem, substituir x, y por $\emptyset$

$$T\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3\times 0 \\ -0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \checkmark$$

Passo 2: Testar regra de homogeneidade

$$\boxed{T(c \times \vec{v}) = c \times T(\vec{v})} \qquad T: \mathbb{R}^2 \to \mathbb{R}^2 \quad T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3x \\ -y \end{bmatrix}$$

Escrever qualquer vetor e escalar

$$\vec{v} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \qquad c = 4$$

$$T(c \times \vec{v}) = T\left(4 \times \begin{bmatrix} 2 \\ 5 \end{bmatrix}\right)$$

$$= T\begin{pmatrix} 8 \\ 20 \end{pmatrix} = \begin{pmatrix} 3\times 8 \\ -20 \end{pmatrix} = \begin{pmatrix} 24 \\ -20 \end{pmatrix}$$

$$c \times T(\vec{v}) = 4 \times T\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right)$$

$$= 4 \times \begin{bmatrix} 3\times 2 \\ -5 \end{bmatrix}$$

$$= 4 \times \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

$$= \begin{bmatrix} 24 \\ -20 \end{bmatrix}$$

$$T(c \times \vec{v}) = c \times T(\vec{v}) \checkmark$$

$$\begin{bmatrix} 24 \\ -20 \end{bmatrix} = \begin{bmatrix} 24 \\ -20 \end{bmatrix}$$

Passo 3. Testar aditividade

$$\boxed{T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})} \qquad T: \mathbb{R}^2 \to \mathbb{R}^2 \quad T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3x \\ -y \end{bmatrix}$$

Escrever quaisquer vetores

$$\vec{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad w = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

$$T(\vec{u} + \vec{v}) = T\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ -2 \end{bmatrix}\right)$$

$$= T\begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 3\times 5 \\ -1 \end{pmatrix}$$

$$= \begin{bmatrix} 15 \\ -1 \end{bmatrix}$$

$$T(\vec{u}) + T(\vec{v}) =$$

$$T(\vec{u}) = T\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = \begin{pmatrix} 3\times 1 \\ -3 \end{pmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

$$T(\vec{v}) = T\left(\begin{bmatrix} 4 \\ -2 \end{bmatrix}\right) = \begin{pmatrix} 3\times 4 \\ -2 \end{pmatrix} = \begin{bmatrix} 12 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ -3 \end{bmatrix} + \begin{bmatrix} 12 \\ -2 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$$

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v}) \checkmark$$

$$\begin{bmatrix} 15 \\ -5 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$$

Conclusão: $T: \mathbb{R}^2 \to \mathbb{R}^2 \quad T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3x \\ -y \end{bmatrix}$ É uma transformação linear.